

Prayas JEE 2025

Maths

DPP: 3

Trigonometric Functions

- Q1** Value of $\sin^2 5^\circ + \sin^2 10^\circ + \sin^2 15^\circ + \dots + \sin^2 90^\circ =$
 (A) 9
 (B) 8
 (C) 7
 (D) $\frac{19}{2}$
- Q2** If $\sin A = \frac{3}{5}$ and $\cos B = \frac{-12}{13}$, where A and B both lie in second quadrant, then find the value of $\sin(A + B)$
 (A) $\frac{65}{56}$
 (B) $-\frac{65}{56}$
 (C) $-\frac{56}{65}$
 (D) $\frac{56}{65}$
- Q3** Value of $\sin 47^\circ + \cos 77^\circ$ is:
 (A) $\cos 17^\circ$
 (B) $\sin 17^\circ$
 (C) $\tan 17^\circ$
 (D) $\cot 17^\circ$
- Q4** Value of $\frac{\sin 5A - \sin 3A}{\cos 5A + \cos 3A}$ is:
 (A) $\tan A$
 (B) $\cot A$
 (C) $\sec A$
 (D) $\operatorname{cosec} A$
- Q5** Range of $f(x) = \frac{1}{4 \cos x - 3}$ is _____.
 (A) $(-\infty, -\frac{1}{7}] \cup [1, \infty)$
 (B) $(-\infty, \infty)$
 (C) $[1, \infty)$
 (D) None of these
- Q6** $\sin(45^\circ + A) \cos(45^\circ - B) + \cos(45^\circ + A) \sin(45^\circ - B) =$
 (A) $\cos(A - B)$
 (B) $\cos(A + B)$
 (C) $\sin(A - B)$
 (D) $\sin(A + B)$
- Q7** If $\cos A = \frac{4}{5}$, $\cos B = \frac{12}{13}$, $\frac{3\pi}{2} < A, B < 2\pi$, then the value of $\cos(A + B)$ is:
 (A) $\frac{53}{27}$
 (B) $-\frac{33}{65}$
 (C) $-\frac{53}{27}$
 (D) $\frac{33}{65}$
- Q8** Suppose that $a \cos \theta = b$ and $c \sin \theta = d$, for θ and some constants a, b, c, d . Then which of the following is true?
 (A) $a^2 c^2 = b^2 c^2 + a^2 d^2$
 (B) $a^2 d^2 = b^2 c^2 + a^2 c^2$
 (C) $b^2 c^2 = a^2 d^2 + a^2 c^2$
 (D) $a^2 b^2 = b^2 c^2 + c^2 d^2$
- Q9** If $a \sec \alpha - c \tan \alpha = d$ and $b \sec \alpha + d \tan \alpha = c$ then
 (A) $a^2 + b^2 = b^2 + d^2$
 (B) $a^2 + d^2 = b^2 + c^2$
 (C) $a^2 + b^2 = c^2 + d^2$
 (D) None of these
- Q10** If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$, then $x^2 + y^2 + z^2$ is
 (A) r
 (B) r^2
 (C) r^3
 (D) r^{-2}



Q11 If $\cos \theta + \sec \theta = -2$, $\theta \in [0, 2\pi]$, then $\sin^8 \theta + \cos^8 \theta$ equal to

- (A) -2 (B) 1
(C) 2^4 (D) 2^5

Q12 If $\sin^2 \theta = \frac{x^2+y^2+1}{2x}$, then x must be

- (A) -3 (B) -2
(C) 1 (D) None of these

Q13 If $\tan \theta = \frac{p}{q}$, then the value of $\frac{p \sin \theta - q \cos \theta}{p \sin \theta + q \cos \theta}$ is

- (A) $\frac{p^2-q^2}{p^2+q^2}$
(B) $\frac{p^2+q^2}{p^2-q^2}$
(C) 0
(D) 1

Q14 If $\sin x + \sin^2 x = 1$, then the value of $\cos^{12} x + 3 \cos^{10} x + 3 \cos^8 x + \cos^6 x - 1$ is equal to

- (A) 0 (B) 1
(C) -1 (D) 2

Q15 If $10 \sin^4 \alpha + 15 \cos^4 \alpha = 6$, then $27 \operatorname{cosec}^6 \alpha + 8 \sec^6 \alpha =$

- (A) 125 (B) 250
(C) 50 (D) 75

Q16 If A is an obtuse angle, then $\frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} + \frac{\sin A}{\sqrt{1+\tan^2 A}} - 2 \tan A \cot A$ is always equal to

- (A) 1 (B) -1
(C) 2 (D) None of these

Q17 The expression $\frac{\tan A}{1-\cot A} + \frac{\cot A}{1-\tan A}$ can be written as

- (A) $\sin A \cos A + 1$
(B) $\sec A \operatorname{cosec} A + 1$
(C) $\tan A + \cot A$
(D) $\sec A + \operatorname{cosec} A$

Q18

$\frac{\tan\left(\frac{\pi}{4}+x\right)}{\tan\left(\frac{\pi}{4}-x\right)}$ is equal to

- (A) $\left(\frac{1+\tan x}{1-\tan x}\right)^2$
(B) $\left(\frac{1-\tan x}{1+\tan x}\right)^2$
(C) $\left(\frac{1+\sin x}{1-\sin x}\right)^2$
(D) $\left(\frac{1-\sin x}{1+\sin x}\right)^2$



Answer Key

Q1 (D)
Q2 (C)
Q3 (A)
Q4 (A)
Q5 (A)
Q6 (A)
Q7 (D)
Q8 (A)
Q9 (C)

Q10 (B)
Q11 (B)
Q12 (C)
Q13 (A)
Q14 (A)
Q15 (B)
Q16 (B)
Q17 (B)
Q18 (A)



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